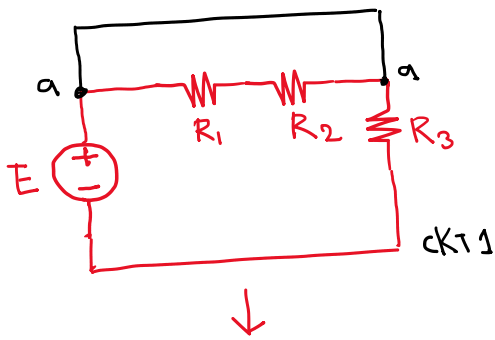
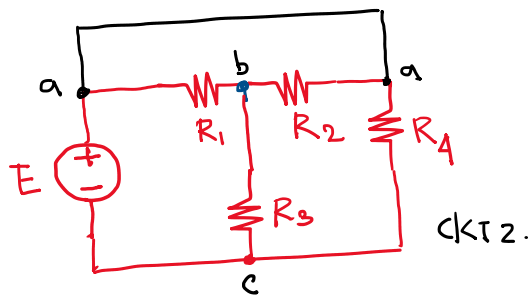
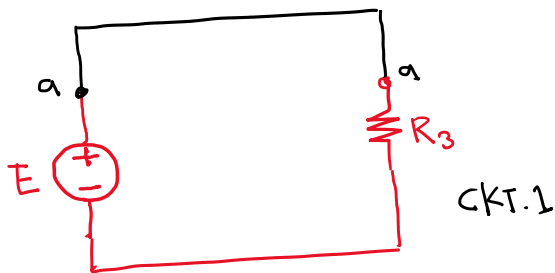




In a circuit, if two ends of a branch are identified as same nodes (a), then the branch is shorted.

In the ckt, branch consists of two resistors R_1 & R_2 . Two ends of the branch are same node (a).

Therefore the branch is shorted



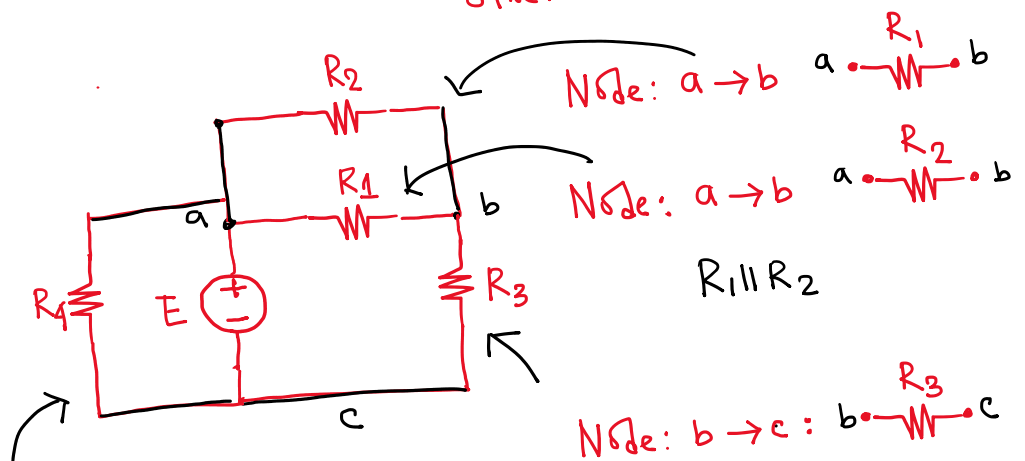
In ckt 2, R_1 and R_2 are not in the same branch. There are two branches

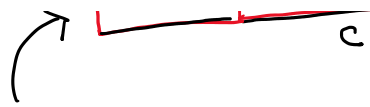


Here, R_1 and R_2 are not shorted as

Branch 1: one end: a
other end: b

Branch 2: one end: b
other end: a

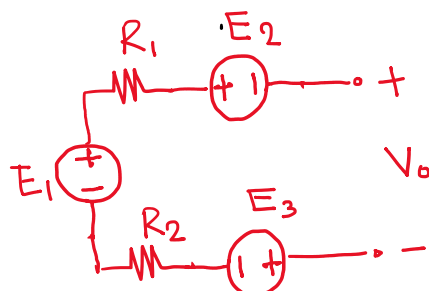
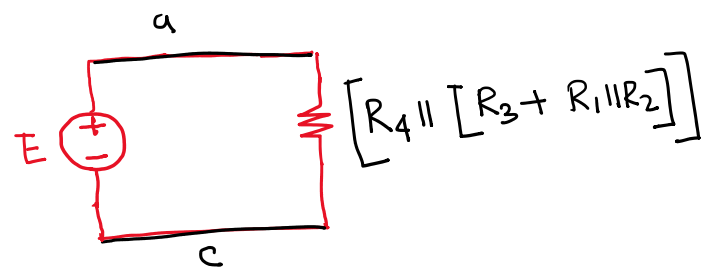
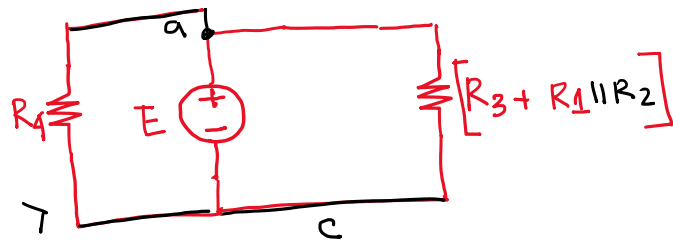
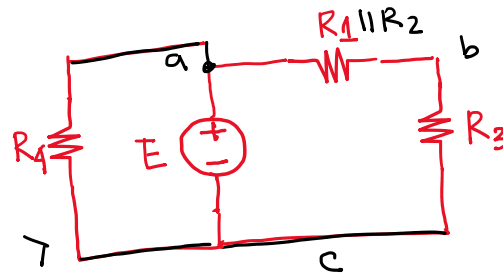




Nbde: $b \rightarrow c$: $b \cdot \text{---} \text{---} \text{---} c$

Nbde: $a \rightarrow c$: $a \cdot \text{---} \text{---} \text{---} R_4 \text{---} c$

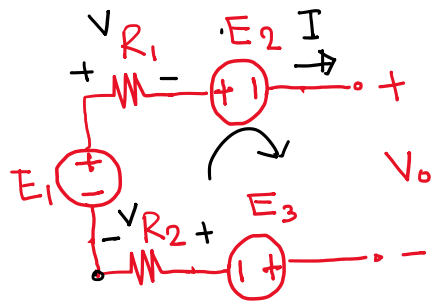
2



Determine V_o .

Make observation: It's an open CKT, so $I = 0$.

Let's assume that the current flowing is I .



Apply KVL:

$$+E_1 - V_{R1} - E_2 - V_o - E_3 - V_{R2} = 0$$

$$\Rightarrow E_1 - IR_1 - E_2 - V_o - E_3 - IR_2 = 0$$

(Applying ohm's law:)

$$\Rightarrow V_o = E_1 - E_2 - E_3 - \cancel{I} (R_1 + R_2)$$

$$\Rightarrow V_o = E_1 - E_2 - E_3$$